Data-Driven Prediction of Foreseeable AI-Training Failure

Location–scale uncertainty, arrival risk, warning horizons and policy effects

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Abstract

This paper formulates foreseeable failure as a probability of entering an independently defined failure region under a stated continuation policy. It separates endpoint risk, cumulative arrival risk, structural change and actionable intervention. Location–scale models, conditional hazards, joint path sets and cost-sensitive warnings provide explicit WPM constructions. Proof trails derive one-sided moment bounds and competing-risk composition, while worked examples show why a safe mean can coexist with nonzero failure probability. The predictor uses observed histories, and its relationship to direct telemetry is described only under the metrics and conditions supported by the archived evidence.

Research lineage and mathematical status. This study develops a WPM mathematical extension with reference to the author's published WPM architecture [1] and foundational n/s scaling-spectrum framework [2]. The n/s paper concerns conditional moment-spectrum parameterization; the WPM paper provides the broader architectural reference. The results proved here are mathematical consequences of stated assumptions, usually applications of standard linear algebra, probability or optimization. The WPM contribution is their organization into an explicit workflow. Modeling constructions and applicability conditions are identified separately; no universal forecasting superiority is asserted.

Predictor inputs. The forecasting layer described here is data driven: it learns from observed histories and matured outcomes. It is not fed the system's true governing equation or unobserved future states at issuance. Physical equations used below define observables, explain a limiting example or specify a separately labeled comparator. Optional supplied-map inverse problems and physics-informed variants must be declared separately; they are not evidence that the default forecast knows the generator.

1 Causal observation and resolution contract

Let $\mathcal{F}_t$ contain the observations actually reported by issuance t: loss measurements, gradient and update statistics, known schedules and verified context. A delayed validation measurement belongs to the information set only when reported. The forecast horizon P is fixed by the query; W selects past extent, and m is a declared temporal sampling or internal-resolution parameter. These variables need not be functions of one another. For each target T, issue a forecast $p_{s,P,r}$

using only $\mathcal{F}_s$, with lens $r = (W, m)$. Once the label has matured, its loss can enter a causal surface

$$\widehat{L}_t(r; T, P) = \frac{\sum_{s \in \mathcal{A}_t} w_s \ell_T(Y^T_{s,P}, p_{s,P,r})}{\sum_{s \in \mathcal{A}_t} w_s}. \tag{1.1}$$

The same matured set $\mathcal{A}_t$ is used in numerator and denominator. Extend maturity when persistence or reporting delay requires follow-up beyond $s + P$. Unresolved negative outcomes are not known negatives. Empty history is missing risk, not zero risk. A pocket is a connected low-risk region under a specified adjacency and threshold. Retain each pocket's flow cross-sections, position, existence, signed errors and identity history. Fixed uses an eligible reference lens; Multi aggregates same-target alternatives; Auto changes a specified policy with causal context and matured utility. A pocket minimum is not an automatic event probability. An interpretable candidate event model is

$$p_{t,P} = \operatorname{logit}^{-1}\left(\beta_0 + \beta^\top z_t + \sum_i \alpha_i \phi(\mathcal{P}_i) + \sum_{i<j} \gamma_{ij} \psi(\mathcal{P}_i, \mathcal{P}_j)\right). \tag{1.2}$$

The coefficients, feature maps and any calibration map must be fitted from permitted data. Pocket labels must not have an artificial ordinal meaning; use a permutation-invariant aggregation or preserve a well-defined relational graph. Shared observations create dependence among pockets and cannot be counted as extra independent measurements. The compact archived AI representation used observable location/scale proxies in its multi-system scenario comparison, not explicit fits of all n/s parameters. A full scaling-spectrum branch is an optional extension that must calculate and qualify its own descriptors. Structural location/scale features and an actually fitted n/s spectrum are not interchangeable descriptions.

2 A failure criterion must be independent of the score

Let Z_t be a monitored state, such as clean validation loss and auxiliary observables. Choose a failure region $\mathcal{B}$ before interpreting predictions. It may represent sustained degradation, nonfinite optimization state, a violated operational constraint or failure to reach a declared quality level within a budget. These are different targets.

Do not define failure as "the WPM score was high." The label must be measurable from independent operational criteria. If the definition requires persistence for d steps, its onset and maturity rules must include those steps. The continuation policy, known schedule, observation clock and reporting delays are also part of the query.

Define first-entry time $\tau_t = \inf\{h \geq 1 : Z_{t+h} \in \mathcal{B}\}$ while the current state is outside the region. The central target is $C_t(P) = \mathbb{P}(\tau_t \leq P \mid \mathcal{F}_t)$. A separate endpoint target is $\mathbb{P}(Z_{t+P} \in \mathcal{B} \mid \mathcal{F}_t)$. A model can answer one accurately and not the other.

3 Location and scale as distinct predictive objects

For scalar Z, define $\mu_{t,P} = \mathbb{E}[Z_{t+P} \mid \mathcal{F}_t]$ and $s^2_{t,P} = \operatorname{Var}(Z_{t+P} \mid \mathcal{F}_t)$. A data-driven model can use separate target-specific lenses for these quantities, or predict event risk directly. A positive scale can be parameterized by $s = e^{g_\eta(z_t, P)}$.

Recent telemetry variability is not automatically predictive variance. Future uncertainty includes process noise, unobserved state, model error and parameter uncertainty. A local Gaussian observation-noise estimate supplies only one component. Structural descriptors can organize

information a simple telemetry comparator discards; they do not create new independent observations when computed from the same recording.

An optional complete n/s branch must actually estimate admissible moment spectra and their parameters. A compact location–scale proxy branch should be named as such. The spectrum equation is not a supplied optimizer evolution equation, and a successful learned response predictor does not prove closed-form temporal evolution of its spectrum parameters.

4 A safe mean can coexist with substantial risk

If the supplied predictive law is $Z \sim N(\mu, s^2)$ and failure is $Z > b$, endpoint probability is $1 - \Phi((b-\mu)/s)$. A mean below b does not make this zero.

Worked example 4.1 (An explicit location–scale risk)**.** Take b=1 and $\mu = 0.8$. At s=0.1, endpoint failure probability is $1-\Phi(2)$; at s=0.2 it is $1-\Phi(1)$. The mean has not changed. Under a non-Gaussian predictive law the same mean and variance need not give these probabilities, so the distributional assumption must remain visible.

An uncertainty envelope touching a boundary is a geometric warning feature, not a unique probability equation. A central interval does not describe an entire path, and its intersection with failure is not equivalent to actual failure. Tail shape and temporal dependence matter.

5 A one-sided bound when only moments are available

Theorem 5.1 (Cantelli bound)**.** *If* $\mathbb{E}Z = \mu$, $\operatorname{Var} Z = \sigma^2 < \infty$ *and* $d>0$, *then*

$$\mathbb{P}(Z - \mu \geq d) \leq \frac{\sigma^2}{\sigma^2 + d^2}. \tag{5.1}$$

Proof. For any a>0, the event $Z - \mu \geq d$ implies $(Z - \mu + a)^2 \geq (d + a)^2$. Markov's inequality gives the bound $(\sigma^2 + a^2)/(d + a)^2$. Minimizing over a gives $a = \sigma^2/d$ and the stated expression. The zero-variance case follows directly. □

The same argument holds conditionally when the conditional moments exist and the assumptions are interpreted given $\mathcal{F}_t$. It is an upper bound, not an estimated exact probability, and can be loose. With $\mu = 0.8$, b=1 and $\sigma = 0.1$, it gives at most $0.01/(0.01 + 0.04) = 0.2$, quite different from the Gaussian tail. Estimated moments also need uncertainty; inserting optimistic estimates does not create a certificate.

6 First-entry risk and coherent warning horizons

Conditional hazards $h_j = \mathbb{P}(\tau = j \mid \tau \geq j, \mathcal{F}_t)$ yield

$$S(P) = \prod_{j=1}^{P} (1 - h_j), \qquad C(P) = 1 - S(P). \tag{6.1}$$

This is the chain rule for survival and implies monotonicity of C in P. It does not require independent future states. Marginal endpoint risks cannot be substituted for hazards without additional assumptions.

For a predeclared horizon grid $\mathcal{P}$, define a warning horizon at probability level q by

$$P_q(t) = \min\{P \in \mathcal{P} : C_t(P) \geq q\}, \tag{6.2}$$

with no-crossing status if the set is empty. This queries a coherent risk curve; it does not optimize an unrestricted P to make the forecast look accurate. If the desired terminal time stays fixed as new observations arrive, the remaining horizon decreases and a refreshed forecast has a new issuance.

Worked example 6.1 (A warning-horizon trail)**.** Hazards $(0.1, 0.2, 0.15)$ give cumulative risks $(0.1, 0.28, 0.388)$. At q=0.3 on the grid $\{1, 2, 3\}$, the first warning horizon is 3. A probability at step 2 alone cannot establish this path-level arrival distribution.

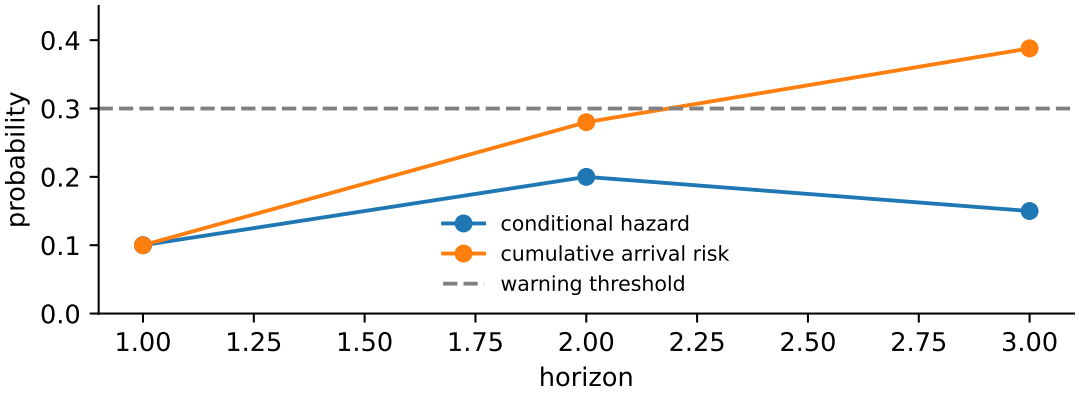


Figure 1: Conditional hazards 0.1, 0.2 and 0.15 imply cumulative arrival probabilities 0.1, 0.28 and 0.388. Hazards and cumulative probabilities have different meanings.

7 Competing failure mechanisms

Suppose failure can occur by several causes k, and $h_{j,k}$ is the conditional probability of first failure at j by cause k given survival through j-1. Assume cause categories at first entry are mutually exclusive, with $\sum_k h_{j,k} \leq 1$. Then

$$S(P) = \prod_{j=1}^{P} \left(1 - \sum_k h_{j,k}\right), \tag{7.1}$$

$$F_k(P) = \sum_{j=1}^{P} S(j-1)h_{j,k}, \qquad S(0) = 1. \tag{7.2}$$

Proof. At each at-risk time, the total first-failure probability is the sum of mutually exclusive cause probabilities. Chain multiplication gives survival. A cause-k failure at j requires survival through j-1 followed by that cause, giving the summand. Sum over disjoint failure times. □

If mechanisms can co-occur, define first-entry categories carefully or use their joint event law. Summing arbitrary marginal probabilities can exceed one. A high gradient-risk score and high data-quality-risk score are not automatically independent causes.

8 Joint path uncertainty and safety statements

Pointwise confidence intervals at several horizons need not provide simultaneous path coverage. If a conditional path set $\mathcal{C}_t$ satisfies $\mathbb{P}(Z_{t+1:t+P} \in \mathcal{C}_t \mid \mathcal{F}_t) \geq 1 - \alpha$ and every path in it stays outside $\mathcal{B}$, then $C_t(P) \leq \alpha$.

Proof. Failure paths cannot lie in $\mathcal{C}_t$ by the safety premise. The failure event is therefore contained in the complement of the path-set event, whose conditional probability is at most α. □

The conditional coverage premise is strong. A marginally calibrated interval or an unverified simulated ensemble does not automatically satisfy it at every issuance. A set intersecting failure does not yield a converse risk lower bound: it can contain failure paths with negligible or unspecified probability.

For joint predictive samples, estimate the arrival event on each complete path, retaining temporal dependence. Multiplying independent-time approximations can misstate risk. Monte Carlo sampling error is distinct from model and calibration error; a large number of samples can estimate a wrong model very precisely.

9 Censoring, stopping and policy changes

For a run observed only through c steps after issuance, no failure by c does not imply no failure by P>c. Survival methods can represent right censoring under their assumptions. A complete-window method can instead omit unresolved windows. Labeling every stopped run healthy creates a systematic bias.

If the monitor triggers early stopping or a learning-rate change, the observed trajectory follows a changed policy. A forecast of failure under no intervention is counterfactual after that action. A later healthy outcome does not prove that the original alert was a false positive for the no-intervention target. Observational correlation between warnings and remedies does not identify remedy effects.

The policy must therefore be part of the deployment contract: predict the currently announced continuation, predict the monitored adaptive policy itself, or estimate counterfactual action responses with justified causal assumptions. These are separate models.

10 Warnings and their costs

For one binary decision with false-warning cost c_A and missed-failure cost c_M, warn when $p > c_A/(c_A + c_M)$. Derive this by comparing $c_A(1 - p)$ with $c_M p$. At equality the two expected costs coincide.

Repeated warnings, action budgets, warning delay and intervention benefit lead to a sequential decision problem. A horizon-specific calibrated probability does not automatically control the probability of any false alert over a full run. Report the warning policy, its reporting cadence and the target population. A high probability of structural change may call for no intervention if the change is benign.

11 An end-to-end application worksheet

Define a clean validation-loss boundary with a stated persistence rule, and keep it independent of the predictor. Choose the initially healthy population and a P-step continuation policy. Fit telemetry, location–scale and combined candidates from histories, using label maturity that includes the confirmation block.

At issuance, build target-specific landscapes from past issued losses, extract pockets and choose a qualified same-target policy. Preserve full flow, position and existence rather than only the lowest-risk cell. Fit scale and event probabilities separately when appropriate. A failure law might be learned directly from causal features or composed from a justified joint path model.

Return the failure definition, horizon, probability or bound, applicability status and the information used. If only moment assumptions support a bound, label it as a bound. If no precursor exists in the information set, no amount of pocket arithmetic can guarantee advance identification. New sensors or context can add information; deterministic transformations of the same history cannot.

12 Closeness to telemetry: the precise supported statement

Scoped evidence statement. In a reported latent-failure comparison, long-horizon location-plus-scale discrimination was close to the specified telemetry comparator: AUROC approximately 0.822 versus 0.819 at that horizon [5]. Other horizons and a later cohort differed, so this is neither statistical equivalence nor universal closeness. The compact multi-system transition, oscillation, instability and plateau comparisons use different labels and proxy features and must not inherit this number [6].

The relevant model is data driven and not supplied the optimizer's true future equations. A physics-style quadratic recurrence can illustrate risk, but it is not the source of the empirical comparison. The observation that a learned representation is competitive with telemetry on one metric does not establish prediction of final accuracy, generalization, arbitrary hardware failure or production language-model behavior.

13 Rules and solved questions

1. Define failure independently of the risk score and distinguish it from benign structural transition.
2. State endpoint, arrival, persistence and policy semantics explicitly.
3. Keep location, predictive scale, measurement noise and model uncertainty distinct.
4. Use conditional hazards or joint paths for arrival risk; interpret coverage at its actual conditioning level.
5. Handle censoring and post-warning interventions without inventing negative labels or counterfactual outcomes.

Exercise 13.1. Can a safe predicted mean and a wide uncertainty model imply nonzero failure risk?

Solution. Yes. The Gaussian example has mean below the boundary and positive tail probability. Envelope intersection alone does not determine that probability.

Exercise 13.2. Does a healthy trajectory after remediation prove a no-intervention forecast was wrong?

Solution. No. The observed outcome is under a different policy. The unobserved no-intervention outcome requires separate causal identification.

Scope of the mathematical exposition

Unless explicitly identified as a sourced evidence statement, numerical values are constructed arithmetic examples. Application requires specifying the observation process, fitting unknown quantities, and checking reliability for the intended population and horizon. This study shares the AI prediction collection DOI 10.5281/zenodo.23784369.

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